

COURSE TITLE : YARN FORMATION I -PRACTICAL
COURSE CODE : 3067
COURSE CATEGORY : B
PERIODS/WEEK : 5
PERIODS/SEMESTER : 75
CREDITS : 3

COURSE OUTCOME

Sl.No	Student will be able to
1	Understand different types of drives & their applications.
2	Set the different setting points on spinning preparatory machines up to the processing of sliver.
3	Diagnose the defects in the machineries and rectify
1	Draw and study the gearing plan of different machines for the process.
2	Determine the draft constants and recommend change wheels for different feed and delivery hanks.

On completion of course the student will be able to understand

1. TRANSMISSION OF MOTIONS & COTTON SELECTION & MIXING

1. Study various belt drive, suitability and functions.
2. Understand speed calculations.
3. Draw the different types of gear drives and explain their application.
4. Suitability of chain drives.
5. Principle of working of friction clutch and brakes.
6. Understand the principle of working of cone drums.
7. Understand the use of helical, bevel and spur wheels
8. Know the use of cams
9. Application of worm & worm wheels
10. Practice hand stack mixing

2. BLOW ROOM

1. Draw the line sketch of modern blow room.
2. Calculate cleaning efficiency of openers in blow room.
3. Draw the gearing diagram of scutcher.
4. Determine draft and production of scutcher.
5. Change the draft wheel for different hank of laps.
6. Check the lap variation by weighing lap.

3. CARDING- SETTING

1. Set the feed part of card for different staple length of cotton
2. Set flats for different staple length of cotton
3. Set the mote knives and licker– in under casing for different cottons.
4. Practice to set transfer roll to doffer
5. Practice to set front plate for different staple cotton
6. Practice to set back plate of card
7. Practice to set taker- in to cylinder.
8. Practice to set doffer to cylinder
9. Run the card and thread the sliver

4. CARDING- CALCULATION

1. Draw the gearing diagram of card
2. Identify the draft change wheel, production change wheel, web tension draft wheel
3. Determine the draft constant and recommend draft change wheel for different hank of sliver
4. Practice changing of draft wheel, Production change wheel and web tension draft change wheel
5. Find out production and efficiency

LIST OF EXPERIMENTS

Belt drives- Plain belt, Cross belt, flat belt and V-belt

Rope drive

Speed calculations using belt drives

Gear drive – types of gears – spur wheels, helical wheels, Bevel Wheels, Worm and Worm Wheels, chain drive- application of each type

Speed calculations using gear wheels

Cone drums drive and its applications

Draw the line sketch of modern Blow room

Calculate the cleaning efficiency of openers

Find out the cleaning efficiency of blow room

Draw the gearing diagram of scutcher

Determine the draft and production of scutcher

Change the draft wheel for different hank of lap

Check the lap variation by weighing the lap

Operate blow room machinery

Set the feed part of the card for different staple length of cotton
Set the flats for different staple length of cotton
Dismantle mote knives, taker – in under grid and re-set
Set the taker– in and doffer to cylinder for different staple of cotton
Set the back plate and front plate to the cylinder for different staples of cotton
Draw gearing diagram of card, identify draft change wheel, tension draft change wheel, production change wheel
Calculate the speeds and draft between various organs.
Determine the draft constant, find out draft change wheel for different hanks of sliver
Change draft wheels for different hank of sliver
Change production change wheel, tension draft wheel

LIST OF EQUIPMENTS/ MACHINERIES

1. Lab Model Blow Room
2. Carding Machine
3. Multiple leaf gauges
4. Spanner set

REFERENCE

1. T K Pattabhiraman - Essential calculation of practical cotton spinning-
2. W S Taggart - Cotton spinning
3. W S Taggart - Cotton spinning calculation